

# Ellen Guo

ellenguo2024@gmail.com [www.linkedin.com/in/ellenguo79/](http://www.linkedin.com/in/ellenguo79/) [Engineering Portfolio](#) 612-720-6898

## Education

### Stanford University | Stanford, CA

2024 - 2028

B.S. in Mechanical Engineering | Relevant Coursework: Chemical Foundations (CHEM31E), Engineering Mechanics–Statics and Dynamics (E14, E15, ME80), Product Realization (ME102/3/4), Thermodynamics (ME30), Fluids (ME70), Heat Transfer (ME131)

### Mounds View High School | Arden Hills, MN + University of Minnesota-Twin Cities | St. Paul, MN

2020 - 2024

GPA: 4.504/4.00 | No. 1 class rank, US Presidential Scholar Candidate, International Science and Engineering Fair Finalist, National Merit Finalist, Shoreview Caring Youth Award, Dual Enrollment at UMN taking 4 CS, comp bio, and literature courses

## Work Experience

### Axon | Boston, MA

June - August 2026

#### Product Management Intern

- Led evaluation of 150+ next-generation product features informing 2027 spending and headcount allocation; 2 approved features projected to save \$600K+ in 2027.
- Built and launched a new incident reporting process, cutting support team administrative work 25% and time until redeployment by 60%; adopted company-wide after sign-off from VP of Product and Senior Global Support leadership.
- Engineered and implemented 2 key product design changes that improved next-gen product tamper resistance by 21%

### Medtronic | Plymouth, MN

June - August 2025

#### New Product Development Engineering Intern

- Developed and conducted design optimization and composite materials testing for next-gen atherectomy product
- Authored and internally published protocol and report for FDA Class III embolic device; results will be used for next-gen coil devices, limited-release launch support, and increasing sales

### Third Party International | Eden Prairie, MN

June - August 2024

#### Automation Consultant

- Learned and utilized VBA/Macros to streamline and automate multiple Excel processes for aerospace manufacturing companies, increasing work efficiency by 200% and removing human error

### Quanex | Arden Hills, MN

June - August 2024

#### Process Engineering Intern

- Designed and cut custom Anderson Windows & Doors products using AutoCAD, CNC programming, and CNC cutting
- Fixed custom in-house fabricated machines from malfunctions during production hours

### Coding & Computing Summer Camp | Arden Hills, MN

#### Founder, Camp Counselor

June - August 2023

- Founded a free camp for 30+ local middle schoolers to inspire interest in coding and technology by teaching them to code animations, websites, and games in Javascript, Python, C++; organized camp logistics and created camp curriculum

## Relevant Extracurricular Activities

### Stanford Student Space Initiative

#### IREC Project Systems Lead

June 2025 - Present

- Created overall rocket design, specifications, budget, and timeline; led subteams, overall integration, and project direction
- Facilitated advanced technical R&D projects such as airbrakes, zero g crystallization, on rocket going 30k ft and mach 2
- Analyzed various unique rocket designs for aerodynamic performance utilizing ANSYS CFD and Jupyter Notebook

#### Propulsion Team and Personal R & D Projects, NAR L1 and L2 HPR Certified

September 2024 - Present

- Constructed 4 high-powered rockets utilizing CF & FG composites, different epoxies, modified electronics, CAD, and 3D printing with various filaments for molds and structures; currently building a subminimum two-stage spaceshot project
- Innovated design of injector + ignitor casings for hybrid rocket via research, testing, and simulations on Ansys and Python

### Stanford Mechanical Engineering Research Lab

#### CHARM Lab Marine Autonomous Robotics

September 2025 - Present

- Ideated and prototyped a hand-sized underwater robot that senses salinity, depth, temperature in varying ocean levels
- Designed PCB and innovated electronics waterproofing solutions to withstand high pressure, erosive marine conditions
- Researched underwater communication and wireless charging, focusing on underwater inductive wireless charging

### Stanford Undergraduates in Mechanical Engineering

#### Academic Development President

September 2024 - Present

- Led quarterly events to guide 60+ Stanford mechanical engineering students in developing their academic pathways, research interests, graduate school options, and building connections between students and professors

## Skills & Interests

**Skills:** 3D CAD (Fusion, Solidworks), PCB Design (KiCAD), FEA, CFD (Ansys), Statistical Analysis (Minitab), DFM, Turning/Lathe, Casting, Welding, Materials Testing, Composites, HTML/CSS, Python, MATLAB, Laser Cutting, 3D Printing, Jupyter Notebook, Excel